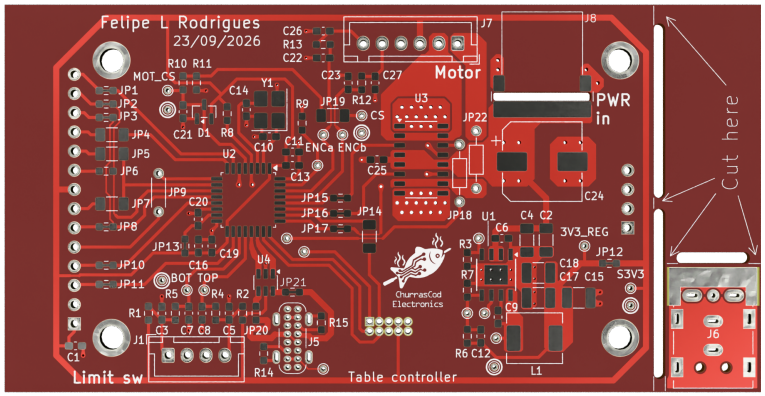
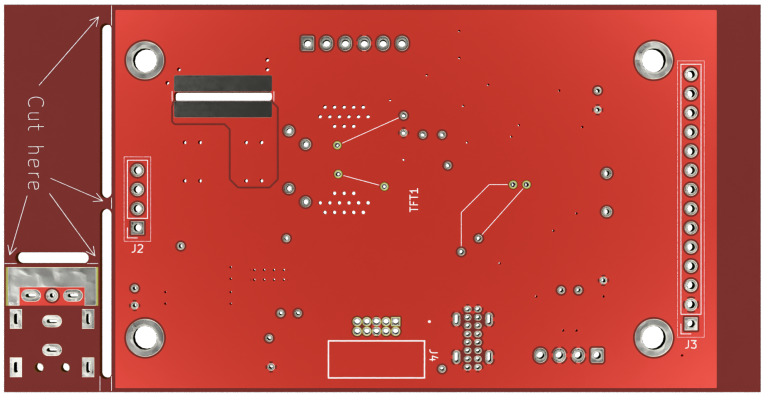
Desk controller fabrication drawing

Top view



Bottom view



ChurrasCod Electronics	**Desk controller**	Sheet: Bare board overview
		File: Desk_controller.kicad_pcb
	Notes	Rev: 1.0.1 / Drwn: FLR
		Mat: Concept / Rev'd: FLR
		Date: 06/10/2026 / Rls'd: FLR
		Dwg: NA / Size: A4
		Generated with KiCad / Page: 1/12

Fabrication requirements

Fabrication requirements

1. FABRICATE PCB IN ACCORDANCE WITH IPC-6012F CLASS 2.

2. FINISHED PCB ACCEPTABILITY PER IPC-A-600M CLASS 2.

3. BASE MATERIAL: FR-4, Tg > 170°C, CAF RESISTANT, UL 94 V-0.

4. LAMINATE MATERIAL SHALL COMPLY WITH IPC-4101F.

5. LAYER COUNT, STACK-UP, BOARD THICKNESS AND COPPER WEIGHTS AS SPECIFIED IN THE STACK-UP DRAWING.

6. COPPER WEIGHTS IN THE STACK-UP REPRESENT STARTING FOIL WEIGHTS. FINISHED CONDUCTOR AND PLATED-HOLE COPPER THICKNESSES SHALL COMPLY WITH IPC-6012F CLASS 2.

7. SOLDER MASK: LPI, BOTH SIDES, COMPLIANT WITH IPC-SM-840E, CLASS H.

8. VIA TENTING SHALL FOLLOW THE SOLDER MASK GERBER DATA. VIA FILLING OR PLUGGING IS NOT REQUIRED UNLESS OTHERWISE SPECIFIED.

9. SURFACE FINISH: LEAD-FREE HASL, PER IPC-6012F.

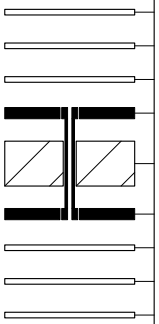
10. PCB MATERIALS AND FINISHES SHALL COMPLY WITH EU RoHS DIRECTIVE 2011/65/EU, AS AMENDED.

11. 100% ELECTRICAL CONTINUITY AND ISOLATION TEST REQUIRED.

12. MATERIAL OR PROCESS SUBSTITUTIONS REQUIRE APPROVAL.

13. THE SUPPLIED GERBER AND NC DRILL FILES SHALL BE THE AUTHORITATIVE SOURCE FOR PCB GEOMETRY, BOARD OUTLINE, COPPER FEATURES AND HOLE LOCATIONS/SIZES. DIMENSIONS AND LAYER IMAGES SHOWN IN THIS DOCUMENT ARE FOR REFERENCE ONLY. IN CASE OF DISCREPANCY, THE FABRICATION DATA SHALL TAKE PRECEDENCE.

Board build up



Material	Layer	Thickness	Dielectric	Type
	F.Paste			Paste Mask
	F.Silkscreen			Legend
	F.Mask	0.01mm		Solder Mask
Copper	**F.Cu**	**0.035mm (1.00oz)**		**Signal**
Core		1.11mm	FR4	Dielectric
Copper	**B.Cu**	**0.035mm (1.00oz)**		**Signal**
	B.Mask	0.01mm		Solder Mask
	B.Silkscreen			Legend
	B.Paste			Paste Mask

Total thickness: 1.2mm

ChurrasCod Electronics

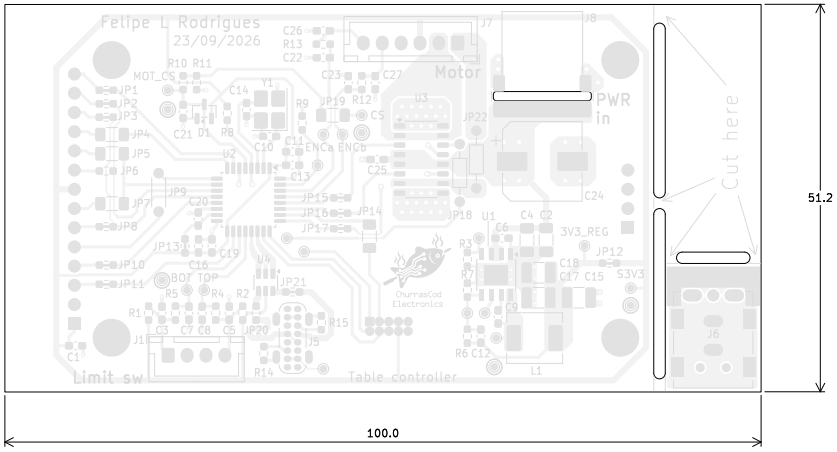
Desk controller

Notes

Sheet: Fabrication requirements
File: Desk_controller.kicad_pcb

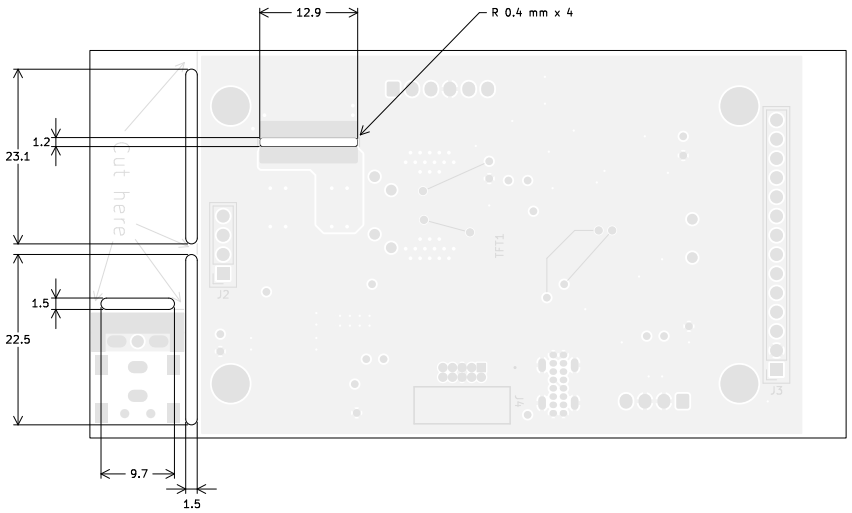
Rev: 1.0.1	Drwn: FLR
Mat: Concept	Rev'd: FLR
Date: 06/10/2026	Rls'd: FLR
Dwg: NA	Size: A4
Generated with KiCad	Page: 2/12

Top fabrication (scale 1:1)



ChurrasCod Electronics	**Desk controller**	Sheet: Top fabrication (scale 1:1)
	Notes	File: Desk_controller.kicad_pcb
		Rev: 1.0.1 / Drwn: FLR
		Mat: Concept / Rev'd: FLR
		Date: 06/10/2026 / Rls'd: FLR
		Dwg: NA / Size: A4
		Generated with KiCad / Page: 3/12

Bottom fabrication (scale 1:1)



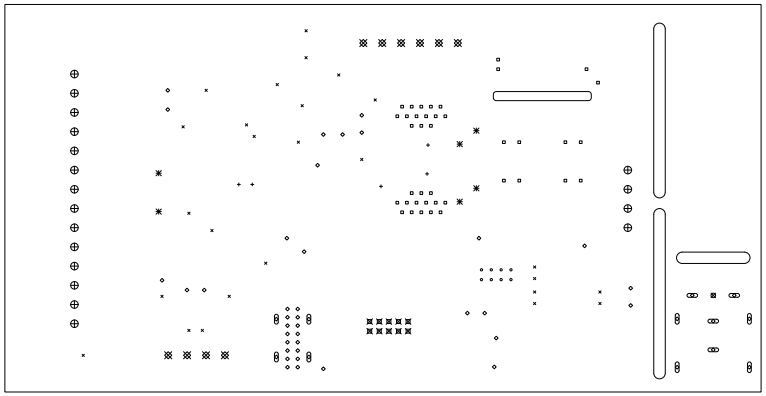
ChurrasCod Electronics

Desk controller

Notes

Sheet: Bottom fabrication (scale 1:1)	
File: Desk_controller.kicad_pcb	
Rev: 1.0.1	Drwn: FLR
Mat: Concept	Rev'd: FLR
Date: 06/10/2026	Rls'd: FLR
Dwg: NA	Size: A4
Generated with KiCad	Page: 4/12

Drill drawing — L1 - L2 (scale 1:1)



Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair
×	26	0.30mm	PTH	Round	F.Cu – B.Cu
○	8	0.30mm	PTH	Round	F.Cu – B.Cu
+	5	0.40mm	PTH	Round	F.Cu – B.Cu
□	40	0.50mm	PTH	Round	F.Cu – B.Cu
◇	37 + 12	0.50mm	PTH	Round + Slot	F.Cu – B.Cu
⊠	1	0.60mm	PTH	Round	F.Cu – B.Cu
✳	6	0.70mm	PTH	Round	F.Cu – B.Cu
⊠	10	0.74mm	PTH	Round	F.Cu – B.Cu
⊠	10	0.95mm	PTH	Round	F.Cu – B.Cu
⊕	18	1.00mm	PTH	Round	F.Cu – B.Cu
	Total 173				

ChurrasCod Electronics

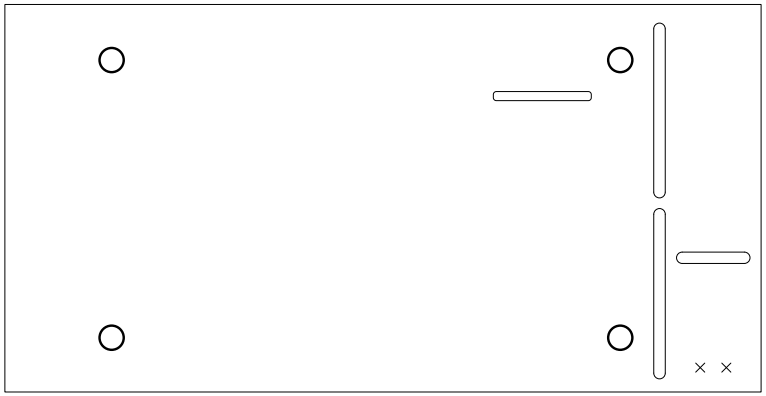
Desk controller

Notes

Sheet: Drill drawing — L1 - L2 (scale 1:1)
File: Desk_controller.kicad_pcb

Rev: 1.0.1	Drwn: FLR
Mat: Concept	Rev'd: FLR
Date: 06/10/2026	Rls'd: FLR
Dwg: NA	Size: A4
Generated with KiCad	Page: 5/12

Drill drawing — L1 - L2 (scale 1:1)



Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair
×	2	1.20mm	NPTH	Round	F.Cu – B.Cu
○	4	3.20mm	NPTH	Round	F.Cu – B.Cu
	Total 6				

ChurrasCod Electronics

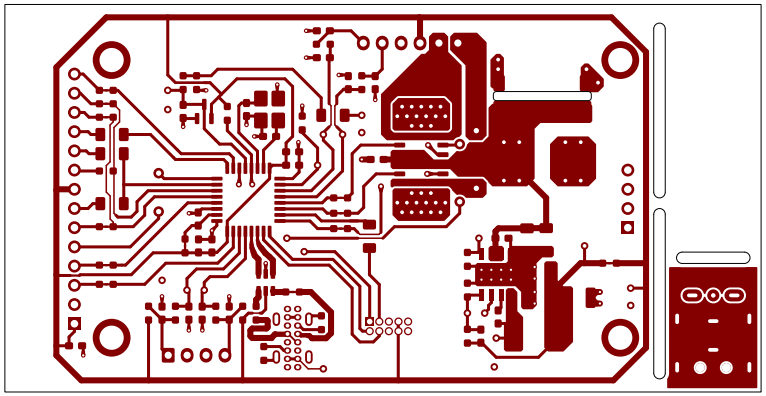
Desk controller

Notes

Sheet: Drill drawing — L1 - L2 (scale 1:1)
File: Desk_controller.kicad_pcb

Rev: 1.0.1	Drwn: FLR
Mat: Concept	Rev'd: FLR
Date: 06/10/2026	Rls'd: FLR
Dwg: NA	Size: A4
Generated with KiCad	Page: 6/12

F.Cu (scale 1:1)



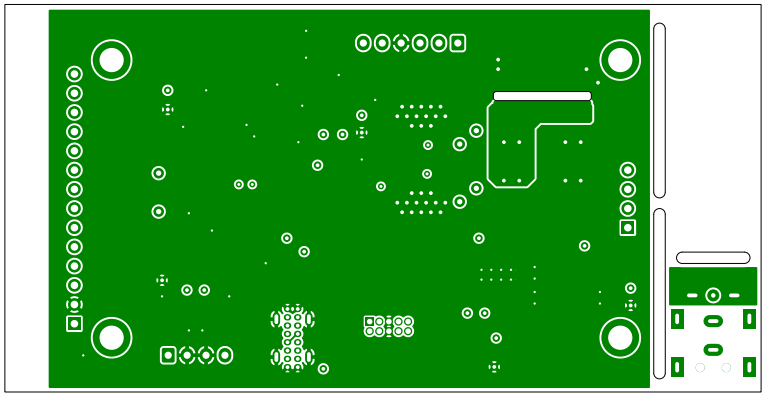
ChurrasCod Electronics

Desk controller

Notes

Sheet: F.Cu (scale 1:1)	
File: Desk_controller.kicad_pcb	
Rev: 1.0.1	Drwn: FLR
Mat: Concept	Rev'd: FLR
Date: 06/10/2026	Rls'd: FLR
Dwg: NA	Size: A4
Generated with KiCad	Page: 7/12

B.Cu (scale 1:1)



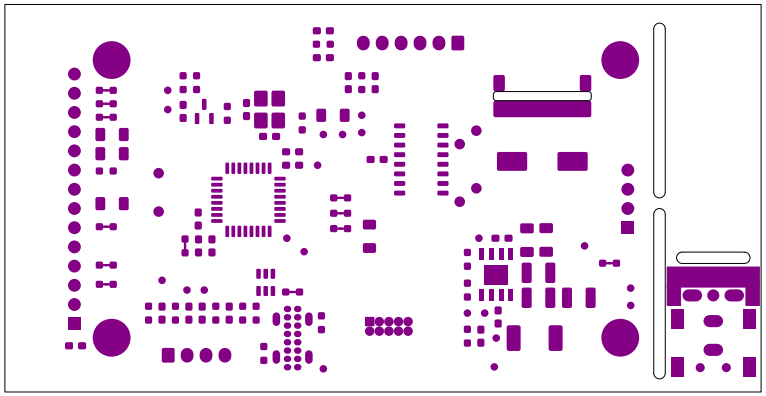
ChurrasCod Electronics

Desk controller

Notes

Sheet: B.Cu (scale 1:1)	
File: Desk_controller.kicad_pcb	
Rev: 1.0.1	Drwn: FLR
Mat: Concept	Rev'd: FLR
Date: 06/10/2026	Rls'd: FLR
Dwg: NA	Size: A4
Generated with KiCad	Page: 8/12

F.Mask (scale 1:1)



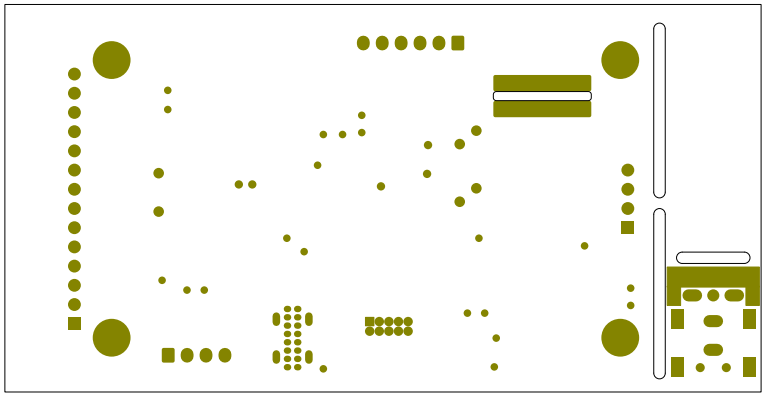
ChurrasCod Electronics

Desk controller

Notes

Sheet: F.Mask (scale 1:1)	
File: Desk_controller.kicad_pcb	
Rev: 1.0.1	Drwn: FLR
Mat: Concept	Rev'd: FLR
Date: 06/10/2026	Rls'd: FLR
Dwg: NA	Size: A4
Generated with KiCad	Page: 9/12

B.Mask (scale 1:1)



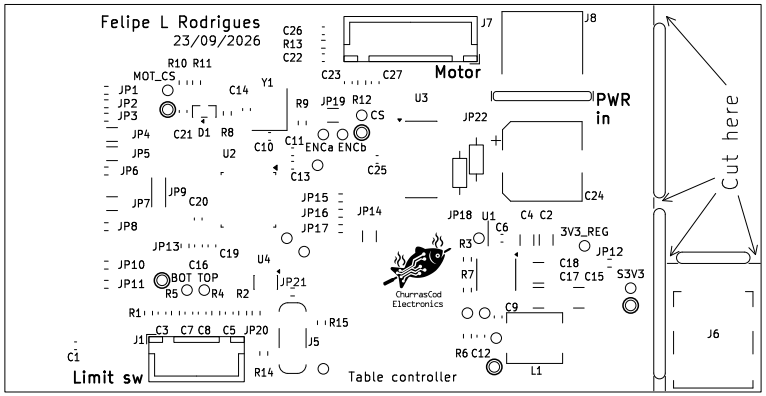
ChurrasCod Electronics

Desk controller

Notes

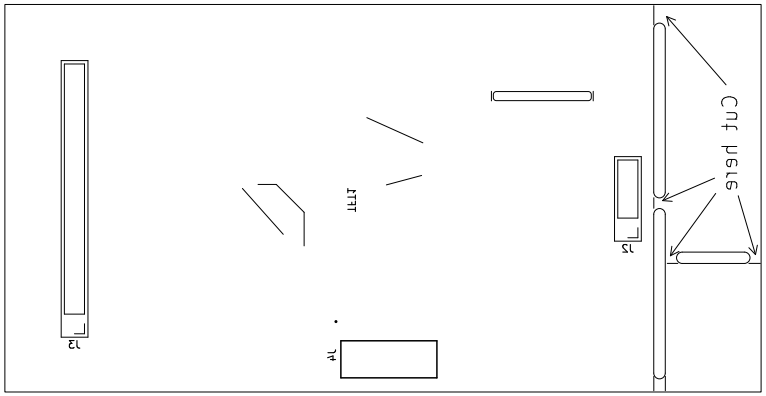
Sheet: B.Mask (scale 1:1)	
File: Desk_controller.kicad_pcb	
Rev: 1.0.1	Drwn: FLR
Mat: Concept	Rev'd: FLR
Date: 06/10/2026	Rls'd: FLR
Dwg: NA	Size: A4
Generated with KiCad	Page: 10/12

F.Silkscreen (scale 1:1)



ChurrasCod Electronics	**Desk controller**	Sheet: F.Silkscreen (scale 1:1)
	Notes	File: Desk_controller.kicad_pcb
		Rev: 1.0.1 / Drwn: FLR
		Mat: Concept / Rev'd: FLR
		Date: 06/10/2026 / Rls'd: FLR
		Dwg: NA / Size: A4
		Generated with KiCad / Page: 11/12

B.Silkscreen (scale 1:1)



ChurrasCod Electronics	**Desk controller**	Sheet: B.Silkscreen (scale 1:1)
	Notes	File: Desk_controller.kicad_pcb
		Rev: 1.0.1 / Drwn: FLR
		Mat: Concept / Rev'd: FLR
		Date: 06/10/2026 / Rls'd: FLR
		Dwg: NA / Size: A4
		Generated with KiCad / Page: 12/12